KSJ RESEARCH

Research Note Vol.5 · Finale

# An Investment Map of AI Energy Assets

Demand → assets → listing → contracts: four volumes on one map — where to invest, where to wait, what to avoid

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2026.11

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# Invest in contracted cash flows (CfD IRR 12.25%, PF min DSCR 1.76x), wait where deals cannot close (PPA room ≤ 17 < 40 charges, B), avoid what leans on prices or multiples (merchant IRR 3.75%)



## Four volumes, four seats, seven verdicts — what separates them is less the return than the size of the assumption that flips it

### Key metrics [*1,*2,*3,*4,*5]

| Key metric | Value | p. |
|---|---|---|
| Invest · KEPCO CfD equity IRR / hurdle | 12.25% / 8% | 3 |
| Invest · DC PF min DSCR / target | 1.76x / 1.45x | 3 |
| CP: min DSCR if power is in the rent | 0.70x | 3 |
| Wait · PPA room vs charges (B) | KRW 5.6–17.0 vs ~KRW 40 | 4 |
| Wait · IPO required vs range discount | 37.9%–43.4% vs 21.9%–35.0% | 4 |
| Wait · REC transitional rule (addendum art. 3, A) | former rules · notice pending | 4 |
| Avoid · merchant equity IRR | 3.75% | 5 |
| Avoid · multiple share 3-name NI / 4-name OP | 39% / 69% | 5 |
| A·D among sourceable inputs (161 in all) | 42 / 145 (29%) | 6 |
| Corrections (git-verified) | 10 (10/10) | 6 |

Source: *1 KSJ Research Vol.1 (Oct 2026) — DART/KRX market-cap split
*2 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
*3 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
*4 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
*5 KSJ Research

Note: Numbers carried over from Vol.1–4 (as of 20 Oct 2026). Verdicts are judgement (J). Volume, page and grade in A2

## Invest

2 assets: Solar equity on a KEPCO CfD · Data-centre PF senior loan

### Contracted cash flows: KEPCO CfD equity IRR 12.25% (hurdle 8%); data-centre PF min DSCR 1.76x (target 1.45x)

- CfD equity: gearing 80% · min DSCR 1.33x — flipped by output: CF 14% → 6.6% [Vol.4 p.4 · J]
- DC PF loan KRW 188.5bn (60% of cost): condition precedent = power billed separately — if included, min DSCR 0.70x on the same debt; re-sized, equity IRR 5.7% [Vol.2 p.7 · J]

## Wait

3 assets: Direct PPA with a data centre · Target Co. IPO · Existing REC-dependent assets

### Returns or rules hold but the deals do not close yet: PPA room KRW 5.6–17.0 < charges ~KRW 40 (B); IPO required discount 37.9%–43.4% > the range's widest 35.0%; existing REC assets keep the former rules (addendum art. 3, A) — notice pending

- Direct PPA (IRR 13.28%): charges must fall by KRW 23 or the buyer must pay that much more — triggers: lower network charges, supply without the grid [Vol.4 p.A1 · J]
- Target Co. IPO (range KRW 66,200–79,500): required discount exceeds the range by 2.9pp–8.4pp — trigger: confirming the margin floor [Vol.3 p.7 · J]
- Existing REC assets: revenue does not vanish with the scheme (addendum art. 3(2)–(3)) — what remains is REC price and demand and the notice's scope; trigger: the notice [Vol.4 p.2 · A]

## Avoid

2 assets: Merchant solar · Listed power-equipment names at ~40x P/E

### They lean on market prices or multiples: merchant IRR 3.75%; multiple contribution 39%–69%

- Merchant: only at the top of the SMP range does it reach 8.2%, barely the hurdle — no RECs [Vol.4 p.4 · J]
- P/E 13.5x → 40.1x: the multiple's share is 39% on 3-name net income vs 69% on 4-name operating income [Vol.1 p.4 · D]

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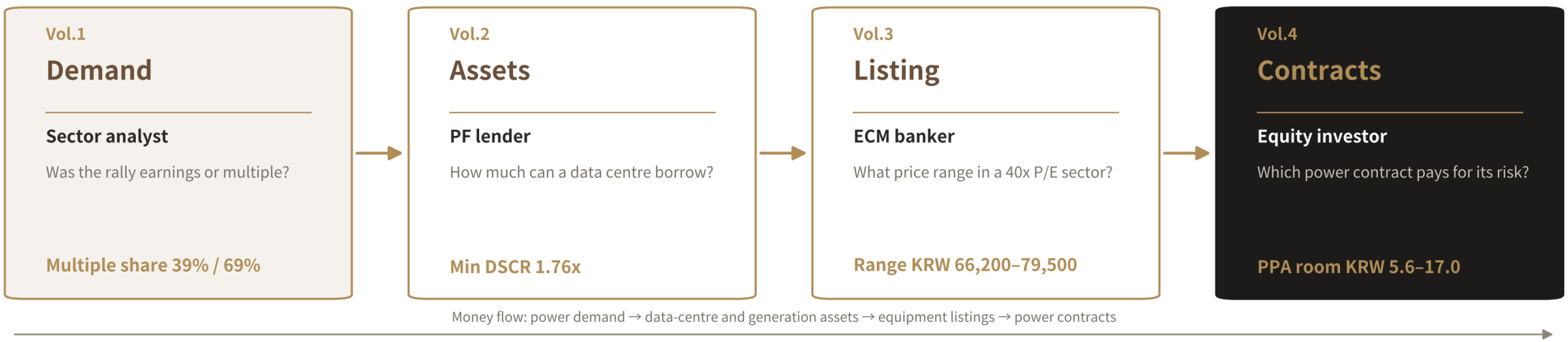

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# Money flow and four lenses: demand (Vol.1) → assets (Vol.2) → listing (Vol.3) → contracts (Vol.4) — one industry, a different question from each seat



## Sector analyst · PF lender · ECM banker · equity investor — the seat each volume took and what it concluded

### The money AI power demand creates passes four turns — each with its own seat and its own decision [*1,*2,*3,*4]




Source: *1 KSJ Research Vol.1 (Oct 2026) — DART/KRX market-cap split
*2 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
*3 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
*4 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
Note: Number at the foot of each box = the volume's key result. Vol.1 measures demand through listed equipment caps; Vol.4 uses a hypothetical 35 MW solar plant

### One-line conclusion per volume — each volume's executive-summary title, verbatim (Korean matched to its build summary 4/4) [*1,*2,*3,*4]

| Vol. · stage | Seat | Question | One-line conclusion |
|---|---|---|---|
| Vol.1 · Demand | **Sector analyst** | Was the rally earnings or multiple? | Demand is real, it shows up on the ground — earnings led the market (3-name net income, 28% of starting cap; multiple-led on 4-name operating income), but P/E also rose from 13.5x to 40.1x |
| Vol.2 · Assets | **PF lender** | How much can a data centre borrow? | Confirming the rent definition (power included or not) is the first diligence item — Korea equity IRR 15.5% (power billed separately) / 5.7% (power included); debt capacity Korea KRW 188bn, U.S. USD 1,232mn |
| Vol.3 · Listing | **ECM banker** | What price range in a 40x P/E sector? | Indicative range KRW 66,200–79,500 (current basis) vs KRW 46,900–57,600 (with peak margins) — the margin floor drives the double-peak risk (FY22A 43.4% vs FY23A 37.9% median discount) |
| Vol.4 · Contracts | **Equity investor** | Which power contract pays for its risk? | The data centre saves 169.8 (energy rate + fuel adjustment + climate charge + fund levy) − required PPA 152.8–164.2 = room 5.6~17.0 — at most ~17 < ~40 network charges in a public example (B) |

Source: *1 KSJ Research Vol.1 (Oct 2026) — DART/KRX market-cap split
*2 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
*3 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
*4 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
Note: The same industry asks different questions from different seats — the map (p2) joins the four answers asset by asset

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# The map: invest 2 · wait 3 · avoid 2 — the size of the flipping assumption matters more than the return

## Base numbers as in each volume · [volume · page · grade] · verdicts and triggers are judgement (J)

### Seven assets: base return · the assumption that flips it and by how much · verdict · trigger [*1,*2,*3,*4,*5]

| Asset (vol. · seat) | Base return · key numbers | Assumption that flips it, and by how much | Verdict (J) | Trigger — what would change it (J) |
|---|---|---|---|---|
| Solar equity on a KEPCO CfD — Vol.4 · Equity investor | Equity IRR 12.25% (hurdle 8%) · gearing 80% · min DSCR 1.33x [Vol.4 p.4 · J] | Capacity factor 16% → 14% (KEEI low) cuts IRR to 6.6% < 8% [Vol.4 p.4 · J] · CAPEX at KEEI top: 8.8% | **Invest** | Re-examine if the contract-market price clears below the KRW 147.7 cap, or P90 output nears 14% |
| Data-centre PF senior loan — Vol.2 · PF lender | Min DSCR 1.76x (target 1.45x) · debt KRW 188.5bn (60% of cost) [Vol.2 p.3 · J] | With power inside the rent, min DSCR 0.70x on the same debt [Vol.2 p.4 · J] — re-sized, debt KRW 100.9bn, equity IRR 5.7% [Vol.2 p.7 · J] | **Invest** | Separate power billing in the lease = condition precedent to drawdown; if unconfirmed, size on the power-included basis |
| Direct PPA with a data centre — Vol.4 · Equity investor | Equity IRR 13.28% (corridor mid KRW 156.5) [Vol.4 p.4 · J] | Buyer's room KRW 5.6–17.0 [Vol.4 p.5 · J] vs public-example charges ~KRW 40 [Vol.4 p.5 · B] — KRW 23 short [Vol.4 p.A1] | **Wait** | Lower network charges (T&D use, settlements) · supply without the grid (notice proviso drops both items) [Vol.4 p.A4 · A] |
| Target Co. IPO — Vol.3 · ECM banker | Range KRW 66,200–79,500 (discount 21.9%–35.0%) [Vol.3 p.6 · J] | Margin floor FY2022A (3.1%) → required 43.4%; FY2023A (5.2%) → 37.9% [Vol.3 p.7 · J] — 2.9pp–8.4pp above the widest 35.0% | **Wait** | Confirm the margin floor: when the next results pin down the trough margin and the required discount falls inside the range (max 35.0%) |
| Merchant solar — Vol.4 · Equity investor | Equity IRR 3.75% (hurdle 8%) [Vol.4 p.4 · J] · no RECs for post-2027 entrants [Vol.4 p.2 · A] | Only SMP held at 148.4 (range top) lifts it to 8.2%, barely clearing [Vol.4 p.4 · J] · even at 2022 actual cost: 7.3% | Avoid | A long fixed-price contract (KEPCO market), or SMP settling at the top of its range |
| Existing REC-dependent assets — Vol.4 · Equity investor | No model — REC article lapses end-2029 (addendum art. 2); holders issued by the deadline and recognised licensees keep the former rules (art. 3(2)–(3)) [Vol.4 p.2 · A] | Revenue does not vanish with the scheme — what remains is REC price and demand, and the notice's scope (qualitative) | **Wait** | The ministerial notice (scope under art. 3(3), counting method under art. 4) and REC prices |
| Listed power-equipment names at ~40x P/E — Vol.1 · Sector analyst | 3-name P/E 13.5x → 40.1x · multiple share 39% (3-name NI) / 69% (4-name OP) [Vol.1 p.4 · D] | The earnings basis decides it (39% vs 69%) · P/E 41.4x → 28.5x means −31% value on the same earnings [Vol.3 p.5 · D] | Avoid | P/E nears the normalised multiple, or earnings catch up (multiple share below 50% on 4-name operating income) |

Source: *1 KSJ Research Vol.1 (Oct 2026) — DART/KRX market-cap split
*2 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
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*4 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
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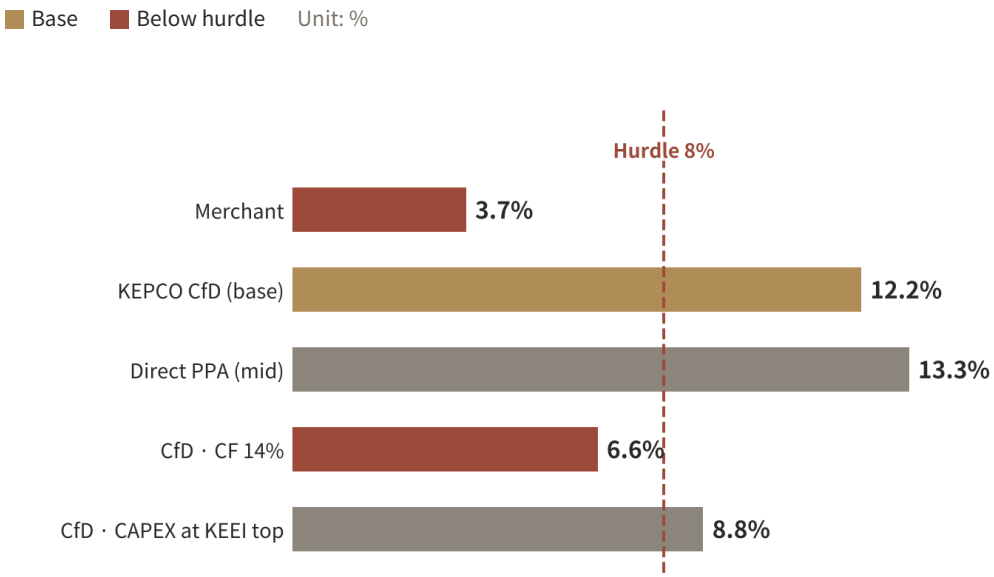
Note: Verdicts and triggers = judgement (J). Rule (J): invest = clears the hurdle or target DSCR and the flipping assumption can be managed by contract terms or CPs (incl. due diligence) · wait = returns (or statutory revenue) hold but closing conditions or a notice are still pending · avoid = misses the hurdle or leans on multiples or market prices (listed equipment: multiple share > 50% on 4-name operating income). [volume · page · grade]: grade = weakest grade among the main inputs (A official > D computed from official > B press/external > C secondary > J judgement); computed values at least D — A2

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# Invest: KEPCO CfD equity (IRR 12.25%) · data-centre PF senior loan (min DSCR 1.76x) — contracted cash flows

## 2 conditions: output for the CfD (P90 study); separate power billing for the PF loan (CP to drawdown)

### KEPCO CfD equity IRR 12.25% clears the 8% hurdle — but a 14% capacity factor cuts it to 6.6% *1



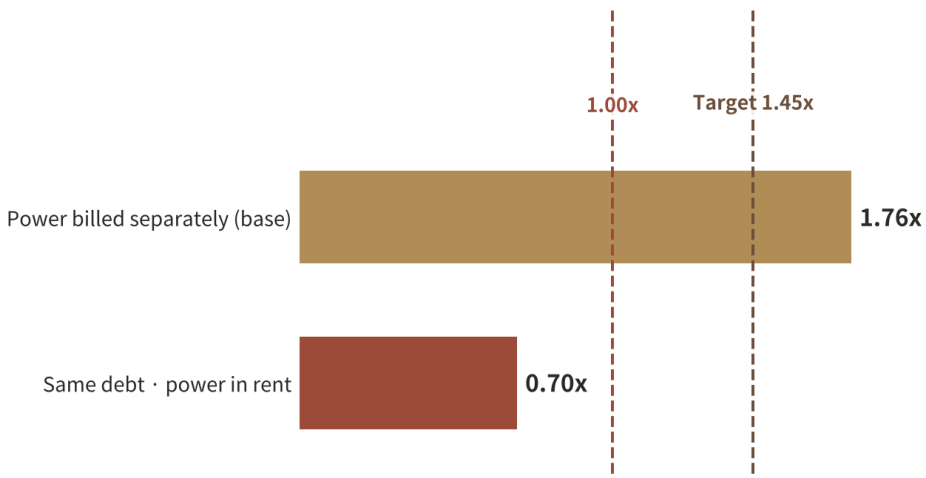

Source: *1 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
Note: Vol.4 p.4 equity IRR and one-at-a-time sensitivities [Vol.4 p.4 · J]. CF range = KEEI (A); CAPEX top = fitted to the KEEI LCOE top

### Why invest: revenue under a 20-year KEPCO contract — 80% gearing still leaves min DSCR 1.33x (target 1.20x) *1

| Item | Value | Source | Grade |
|---|---|---|---|
| Equity IRR · KEPCO CfD | 12.25% | Vol.4 p.4 | J |
| Equity hurdle | 8% | Vol.4 p.4 | J |
| Gearing · KEPCO CfD | 80% | Vol.4 p.4 | J |
| Min DSCR · KEPCO CfD | 1.33x | Vol.4 p.4 | J |
| Target DSCR · KEPCO CfD | 1.20x | Vol.4 p.2 | J |
| KEPCO contract price (2026 round-1 cap) | KRW 147.7 | Vol.4 p.3 | J |
| Capacity factor (base) | 16% | Vol.4 p.1 | A |
| KEPCO CfD equity IRR · CF at range low | CF 14% → 6.6% | Vol.4 p.4 | J |
| KEPCO CfD equity IRR · CAPEX at KEEI top | 8.8% | Vol.4 p.4 | J |

Source: *1 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
Note: Conditions (J): fix gearing after a P90 resource study · recompute if the auction clears below the cap. KEPCO price (J) = the 2026 cap as a proxy for the 2027 market

### Data-centre PF min DSCR 1.76x — with power inside the rent, 0.70x on the same debt *1




Source: *1 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
Note: Vol.2 Korean 40 MW case A — base [Vol.2 p.3 · J], power-included stress [Vol.2 p.4 · J]. Target DSCR is judgement (J)

### Condition precedent: power billed separately in the lease — otherwise capacity KRW 100.9bn, equity IRR 5.7% *1

| Item | Value | Source | Grade |
|---|---|---|---|
| Debt capacity | KRW 188.5bn | Vol.2 p.3 | J |
| Gearing (of total cost) | 60% | Vol.2 p.3 | J |
| Min DSCR (power billed separately) | 1.76x | Vol.2 p.3 | J |
| Target DSCR (sizing) | 1.45x | Vol.2 p.3 | J |
| Equity IRR · power billed separately | 15.5% | Vol.2 p.7 | J |
| Min DSCR · same debt, power in rent | 0.70x | Vol.2 p.4 | J |
| Debt capacity · re-sized with power in rent | KRW 100.9bn | Vol.2 p.7 | J |
| Equity IRR · power included in rent | 5.7% | Vol.2 p.7 | J |

Source: *1 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
Note: Lender's view: contracted rent services the debt — if power sits inside the rent, part of it flows to KEPCO and DSCR drops below 1x

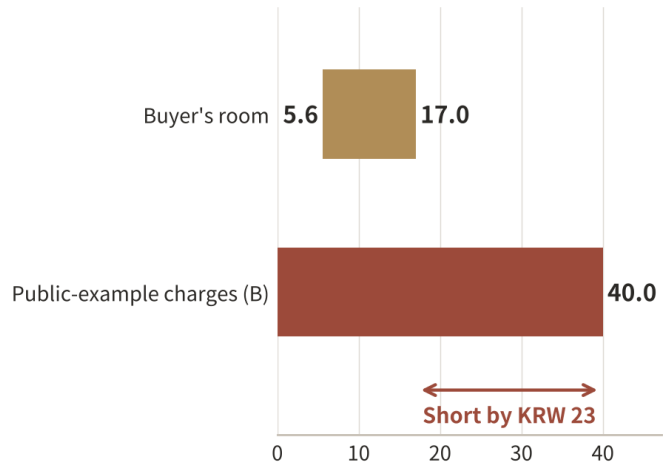
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# Wait: direct PPA (IRR 13.28%, room 17 < 40) · Target Co. IPO (required 37.9%–43.4%) · REC-dependent assets (former rules kept, notice pending) — not the return, the closing conditions

## Triggers move them to invest: network charges for the PPA; the margin floor for the IPO; the ministerial notice for REC assets

### Direct PPA: buyer's room KRW 5.6–17.0 < public-example charges ~KRW 40 (B) — KRW 23 short [*1]

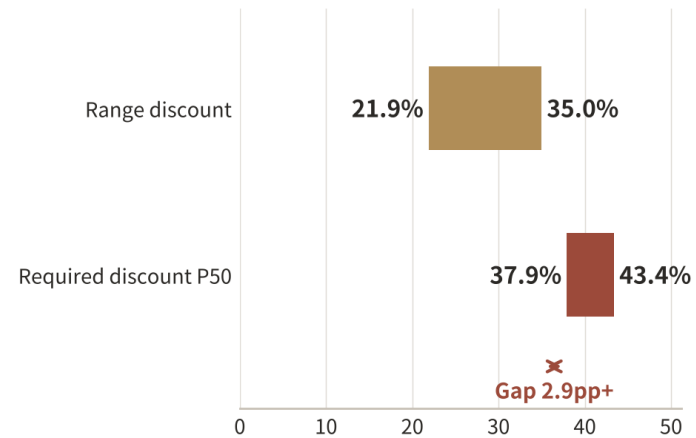
Unit: KRW/kWh




Source: *1 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
Note: Vol.4 p.5 [Vol.4 p.5 · J] · charges = KEPCO-brokered third-party PPA, solar Case 1 (SFOC 2021) [Vol.4 p.5 · B] — direct-PPA charges may be lower (Vol.4 A4)

### Target Co. IPO: required discount 37.9%–43.4% > range discount 21.9%–35.0% — the range is high until the margin floor is known [*1]

Unit: %




Source: *1 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
Note: Vol.3 p.6 range [Vol.3 p.6 · J] · p.7 double-peak (triangular margin) Monte Carlo [Vol.3 p.7 · J]. Gap 2.9pp–8.4pp

### Existing REC-dependent assets: existing holders and recognised licensees keep the former rules (addendum art. 3(2)–(3), A) — wait for the notice [*1]

- The REC article (12-7) stays in force to end-2029; the issuance deadline is 31 Dec 2026 (RE Act addendum arts. 2, 3(1)) [Vol.4 p.2 · A]
- Holders issued RECs by the deadline keep issuing and trading under the former rules, as do pre-act licensees recognised by ministerial notice (addendum art. 3(2)–(3)) — 'the scheme ends, so the revenue goes' does not hold [Vol.4 p.2 · A]
- What remains (J): REC price and demand, and the notice's scope — even counting excess RECs toward targets (addendum art. 4) awaits a notice. Qualitative, no model
- Trigger (J): the ministerial notice (scope under art. 3(3), counting method under art. 4) and REC prices — then value the asset from remaining issuance years and REC revenue share

Source: *1 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity

### Triggers: lower network charges · supply without the grid — no grid use drops T&D and settlement charges (notice proviso, A) [*1]

| Item | Value | Source | Grade |
|---|---|---|---|
| Equity IRR · direct PPA | 13.28% | Vol.4 p.4 | J |
| Direct PPA price (corridor mid) | KRW 156.5 | Vol.4 p.4 | J |
| Seller's required PPA price (per kWh) | KRW 152.8–164.2 | Vol.4 p.i | J |
| Buyer's room for network charges (per kWh) | **KRW 5.6–17.0** | Vol.4 p.5 | J |
| Public example of charges (KEPCO-brokered 3rd-party PPA) | ~KRW 40 | Vol.4 p.5 | B |
| Charge cut (or buyer premium) needed to close | **KRW 23** | Vol.4 p.A1 | J |

Source: *1 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
Note: Notice proviso: items 2 and 4 drop out if the T&D grid is not used [Vol.4 p.A4 · A]. Line that changes the verdict (J): charges ≤ the buyer's maximum room 17.0

### Trigger: confirm the margin floor — when the next results pin down the trough and the required discount falls inside the range [*1]

| Item | Value | Source | Grade |
|---|---|---|---|
| Indicative range (current basis) | KRW 66,200–79,500 | Vol.3 p.6 | J |
| Value per share | KRW 101,810 | Vol.3 p.6 | J |
| Range discount (to value) | 21.9%–35.0% | Vol.3 p.A2 | D |
| 3-name net margin FY2025A | 10.3% | Vol.3 p.2 | D |
| Net-margin floor FY2022A / FY2023A | 3.1% / 5.2% | Vol.3 p.7 | D |
| Required discount P50 · margin floor FY2022A (3.1%) | **43.4%** | Vol.3 p.7 | J |
| Required discount P50 · margin floor FY2023A (5.2%) | **37.9%** | Vol.3 p.7 | J |

Source: *1 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
Note: ECM view (J): before cutting the range, evidence the margin floor (backlog, pricing power) — even an FY2023A floor leaves a gap

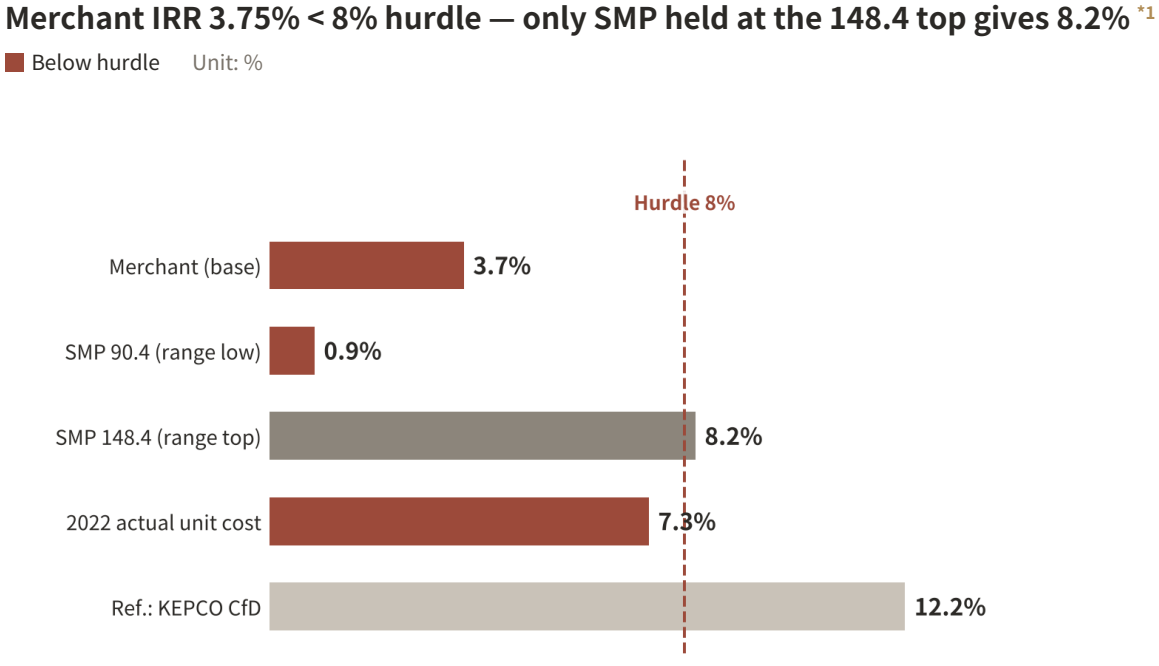
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# Avoid: merchant (IRR 3.75%) · chasing ~40x P/E (multiple share 39%–69%) — returns hang on prices no contract can fix



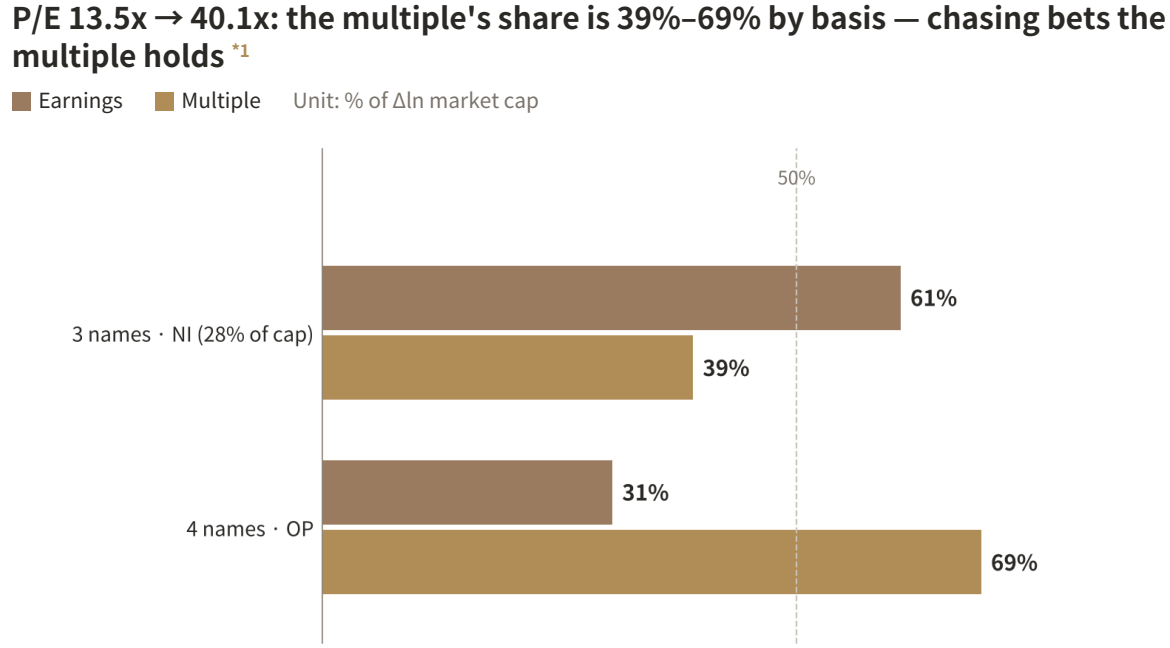
## Common thread: returns hang on prices no contract can fix (SMP, multiples)

### Merchant IRR 3.75% < 8% hurdle — only SMP held at the 148.4 top gives 8.2% *1




Source: *1 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
Note: Vol.4 p.4 [Vol.4 p.4 · J] · SMP range = EPSIS 12-month mainland (A) · no RECs for post-2027 entrants [Vol.4 p.2 · A]

### P/E 13.5x → 40.1x: the multiple's share is 39%–69% by basis — chasing bets the multiple holds *1




Source: *1 KSJ Research Vol.1 (Oct 2026) — DART/KRX market-cap split
Note: Vol.1 p.4 [Vol.1 p.4 · D] — Doosan Enerbility (FY2022A net loss) cannot be split on net income. Normalised P/E [Vol.3 p.5 · D]

### Why avoid, and what would bring them back — all 2 lean on prices nobody can contract *1,*2,*3

| Asset | Why avoid | Signal to revisit (J) |
|---|---|---|
| Merchant solar | IRR 3.75% < 8%, no RECs | A long fixed-price contract |
| Listed power-equipment names at ~40x P/E | Multiple share 39%–69% · −31% if normalised | P/E near the normalised 28.5x |

Source: *1 KSJ Research Vol.1 (Oct 2026) — DART/KRX market-cap split
*2 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
*3 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
Note: Avoid = not at today's price, not never (J). Listed names are a sector-level view, not a call on any stock

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# How the numbers were kept: 42 of 145 inputs that could have a source (29%) are official or computed (A·D) — all 10 corrections kept as commits

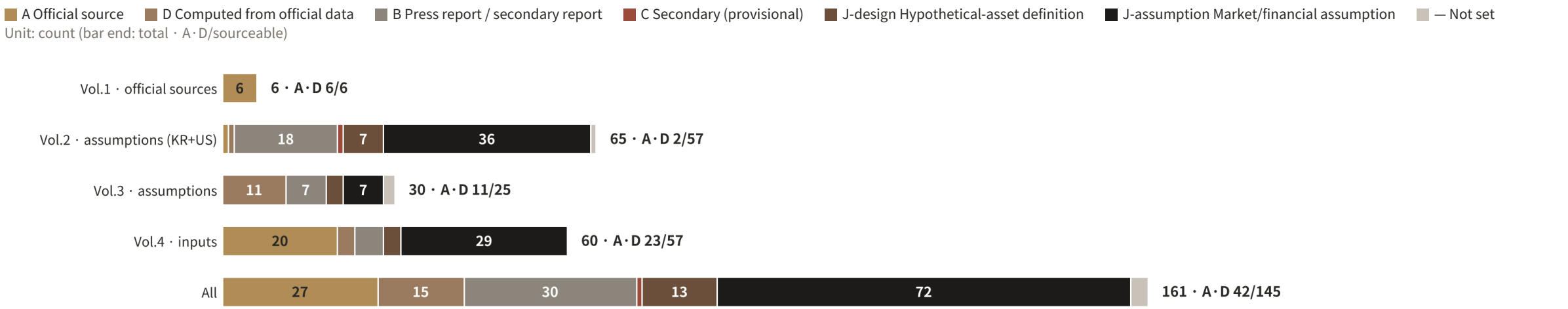
## Corrections by volume (git and build summaries, before → after) · source grades of Vol.1–4 inputs (J split into definitions and assumptions)

### 10 corrections: independent reviews after publishing found assumptions that changed conclusions — kept on record, before → after (git check 10/10) *1

| Vol. | Commit · date | What changed | Before | After | git check |
|---|---|---|---|---|---|
| Vol.1 | ba39c97 · 2026-10-24 | Scope of the 'earnings-led' verdict | 61% of the gain is earnings (3-name basis only) | 3-name NI: earnings 61% (28% of starting cap) · 4-name OP: multiple 69% | match |
| Vol.2 | 67666f4 · 2026-10-22 | Base-case power definition (to pass-through), sourced CAPEX | Base = power included: debt KRW 100.9bn · IRR 7.1% (18.7% if billed separately) | Base = billed separately: debt KRW 204.9bn · IRR 16.1% vs 5.7% with power included | match |
| Vol.2 | a85e0d7·ef6d17a · 2026-10-24 | Gearing cap on CAPEX only | Debt KRW 204.9bn (65%) · min DSCR 1.63x · IRR 16.1% | Debt KRW 188.5bn (60%) · min DSCR 1.76x · IRR 15.5% | match |
| Vol.3 | 5985ffc · 2026-10-22 | Double-peak (peak-margin) risk added | Range KRW 79,500–89,400 (16–25% discount) | Peak-margin range KRW 46,900–57,600 shown alongside | match |
| Vol.3 | 9e33886 · 2026-10-23 | Discount defined as 1 − range ÷ value | Range KRW 67,600–80,400 | Range KRW 66,200–79,500 | match |
| Vol.3 | 7acc942 · 2026-10-24 | What the double-peak verdict depends on | The discount alone cannot cover the double peak | Margin floor decides: 43.4% required at FY2022A · 37.9% at FY2023A | match |
| Vol.4 | a90eb3a · 2026-10-24 | Buyer ceiling = solar-hour avoided rate | Average tariff 181.9 · room KRW 18–29 | Avoided rate 151.3 · room −12.9 to −1.4 (closed before charges) | match |
| Vol.4 | 5bb4cdd·5f599aa · 2026-10-24 | Ceiling in 3 stages (+fuel adj. +climate charge) | Ceiling 151.3 · room −12.9 to −1.4 · PPA IRR 11.6% | Ceiling 165.3 · room 1.1–12.6 · PPA IRR 13.3% | match |
| Vol.4 | 90a9223 · 2026-10-24 | Fund levy on a like-for-like basis with the 40 example | Room 1.1–12.6 (levy not matched) | Room basis 169.8 · room 5.6–17.0 | match |
| Vol.5 | 5d61216 · 2026-10-29 | Legacy REC verdict basis: press report → statute original | Avoid — REC spot market until end-2029 (press, B); transitional rules unchecked | Wait — REC article lapses end-2029; existing holders and recognised licensees keep the former rules (RE Act addendum art. 3(2)–(3), A) | match |

Source: *1 Repository git history (commit hashes) · each volume's build summary before/after
Note: git check = code confirms the 'before' text leaves and the 'after' text appears in the volume's build summary between the first commit's parent and the last commit. Numbers are as published at each point

### 161 inputs: A 27 · D 15 · B 30 · C 1 · J-design 13 · J-assumption 72 · not set 3 — excluding hypothetical-asset definitions, 145 inputs could have a source; J-assumption is 50% of them, hence a flipping assumption on every verdict *1,*2,*3,*4




Source: *1 KSJ Research Vol.1 (Oct 2026) — DART/KRX market-cap split
*2 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
*3 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
*4 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity

Note: Counted in code. Vol.4 = grades in its input file · Vol.1 = official raw-data sources (filings, public bodies, IEA) A · Vol.2–3 = assumption status mapped (data/evidence → D, sourced → A if an official domain else B, verify → B, judgement/neutral → J-assumption, design → J-design). J-design = values that define the hypothetical asset, company or deal (size, timing, structure; Vol.4: 'author's design' in the source field) — no source can exist, so excluded from the base. Base = inputs with a value − J-design. Mapping rules are judgement (J)

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# IC memo: conditional yes on 2 assets — P90 study for the CfD equity, separate power billing for the PF loan; the rest wait for triggers



## Recommendation · key risks · CPs · monitoring (illustrative — hypothetical assets and company)

### Recommendation (judgement, J) *1

| Asset | Verdict | Recommendation |
|---|---|---|
| Solar equity on a KEPCO CfD | Invest | Invest as the base case — fix gearing after a P90 resource study |
| Data-centre PF senior loan | Invest | Approve — separate power billing as a CP to drawdown |
| Direct PPA with a data centre | Wait | Hold — resubmit once charges and the buyer's RE100 premium are confirmed |
| Target Co. IPO | Wait | Hold the range — re-price once the margin floor is evidenced |
| Merchant solar | Avoid | Do not invest — misses the hurdle without a long contract |
| Existing REC-dependent assets | Wait | Hold — check licence date and recognition; resubmit once the notice is out |
| Listed power-equipment names at ~40x P/E | Avoid | Do not chase — revisit near the normalised multiple |

Source: *1 KSJ Research

### Key risks, in numbers *1,*2,*3,*4

| Risk | Impact | Source |
|---|---|---|
| Lower output | CF 14% → 6.6% | [Vol.4 p.4] |
| Power inside the rent | Min DSCR 0.70x on the same debt | [Vol.2 p.4] |
| Charges > buyer's room | KRW 5.6–17.0 vs ~KRW 40 | [Vol.4 p.5] |
| Peak margins (double peak) | Required discount 43.4% | [Vol.3 p.7] |
| SMP fall | SMP 90.4 → 0.9% | [Vol.4 p.4] |
| REC prices, scope of the notice | Former rules kept (art. 3(2)–(3)) — scope and method await a notice | [Vol.4 p.2] |
| Multiple normalises | P/E 41.4x → 28.5x → value −31% | [Vol.3 p.5] |

Source: *1 KSJ Research Vol.1 (Oct 2026) — DART/KRX market-cap split
*2 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
*3 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
*4 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity

### Conditions precedent (CP) *1,*2,*3

| Condition | Applies to | Basis |
|---|---|---|
| Separate power billing in the lease | Data-centre PF senior loan | [Vol.2 p.7] |
| P90 resource study (CF above 14%) | Solar equity on a KEPCO CfD | [Vol.4 p.4] |
| Auction clearing price (assumed KRW 147.7) | Solar equity on a KEPCO CfD | [Vol.4 p.3] |
| Buyer's all-in price at current T&D and settlement rates | Direct PPA with a data centre | [Vol.4 p.A1] |
| Termination payment, parent guarantee | Direct PPA with a data centre | [Vol.4 p.A1] |
| Next results: net-margin floor | Target Co. IPO | [Vol.3 p.7] |
| REC issuance before 31 Dec 2026, pre-act licence, recognition by notice | Existing REC-dependent assets | [Vol.4 p.2] |

Source: *1 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
*2 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
*3 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity

### Monitoring — levels that change a verdict (J) *1,*2

| Indicator | Now | Level that changes it (J) |
|---|---|---|
| KEPCO contract-market price | KRW 147.7 | Below the cap → recompute CfD IRR |
| Direct-PPA charges | ~KRW 40 | ≤ 17.0 → PPA closes |
| SMP (12-month mainland) | KRW 113.2 | Sustained ≥ 148.4 → revisit merchant |
| 3-name net margin | 10.3% | A floor that brings the required discount to ≤ 35.0% — even FY2023A 5.2% gives 37.9% |
| Equipment P/E | 41.4x | Near the normalised 28.5x → revisit |
| KTB 3-yr | 3.84% (2026-10-20) | Rate moves → recompute CfD and PF |
| REC transitional notice, REC prices | Act (A), notice pending | Notice issued (scope, art. 4 method) and prices → re-assess existing assets |

Source: *1 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
*2 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity
Note: 'Now' = each volume's as-of date (20 Oct 2026). Levels come from the volumes; using them as verdict lines is judgement

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# Source map: every result traced to its volume, page and source grade

## Values loaded from each volume's calculation (as of 20 Oct 2026) — no invented numbers

### 46 results → volume, page, grade: page check 46/46 · build-summary check 28/28 *1,*2,*3,*4

| Result | Value | Source | Grade (weakest input) | Check |
|---|---|---|---|---|
| Multiple contribution · 3-name net income | 39% | Vol.1 p.4 | D | O O |
| Earnings contribution · 3-name net income | 61% | Vol.1 p.4 | D | O O |
| 3-name split's share of starting cap | 28% | Vol.1 p.4 | D | O O |
| Multiple contribution · 4-name operating income | 69% | Vol.1 p.4 | D | O O |
| 3-name P/E (start-2023 → now) | 13.5x → 40.1x | Vol.1 p.4 | D | O O |
| Min DSCR (power billed separately) | 1.76x | Vol.2 p.3 | **J (Target DSCR (Korea))** | O O |
| Target DSCR (sizing) | 1.45x | Vol.2 p.3 | **J (Target DSCR (Korea))** | O · |
| Debt capacity | KRW 188.5bn | Vol.2 p.3 | **J (Target DSCR (Korea))** | O O |
| Gearing (of total cost) | 60% | Vol.2 p.3 | **J (Target DSCR (Korea))** | O O |
| Equity IRR · power billed separately | 15.5% | Vol.2 p.7 | **J (Target DSCR (Korea))** | O O |
| Equity IRR · power included in rent | 5.7% | Vol.2 p.7 | **J (Target DSCR (Korea))** | O O |
| Debt capacity · re-sized with power in rent | KRW 100.9bn | Vol.2 p.7 | **J (Target DSCR (Korea))** | O O |
| Min DSCR · same debt, power in rent | 0.70x | Vol.2 p.4 | **J (Target DSCR (Korea))** | O · |
| Indicative range (current basis) | KRW 66,200–79,500 | Vol.3 p.6 | **J (Revenue FY2025A)** | O O |
| Range discount (to value) | 21.9%–35.0% | Vol.3 p.A2 | D | O O |
| Value per share | KRW 101,810 | Vol.3 p.6 | **J (Revenue FY2025A)** | O O |
| Required discount P50 · margin floor FY2022A (3.1%) | 43.4% | Vol.3 p.7 | **J (1-yr net margin distribution)** | O O |
| Required discount P50 · margin floor FY2023A (5.2%) | 37.9% | Vol.3 p.7 | **J (1-yr net margin distribution)** | O O |
| Net-margin floor FY2022A / FY2023A | 3.1% / 5.2% | Vol.3 p.7 | D | O · |
| 3-name net margin FY2025A | 10.3% | Vol.3 p.2 | D | O · |
| Applied P/E → normalised P/E | 41.4x → 28.5x | Vol.3 p.5 | D | O O |
| Equity hurdle | 8% | Vol.4 p.4 | **J (Equity hurdle)** | O · |
| Equity IRR · KEPCO CfD | 12.25% | Vol.4 p.4 | **J (KEPCO contract price (2026 round-1 cap))** | O O |

### Grades: A official · D computed from official · B press/external · C secondary · J judgement

| Result | Value | Source | Grade (weakest input) | Check |
|---|---|---|---|---|
| Equity IRR · direct PPA | 13.28% | Vol.4 p.4 | **J (KEPCO contract price (2026 round-1 cap))** | O O |
| Equity IRR · merchant | 3.75% | Vol.4 p.4 | **J (Max gearing · merchant)** | O O |
| Gearing · KEPCO CfD | 80% | Vol.4 p.4 | **J (KEPCO contract price (2026 round-1 cap))** | O O |
| Min DSCR · KEPCO CfD | 1.33x | Vol.4 p.4 | **J (KEPCO contract price (2026 round-1 cap))** | O O |
| Target DSCR · KEPCO CfD | 1.20x | Vol.4 p.2 | **J (Target DSCR · CfD)** | O · |
| KEPCO CfD equity IRR · CF at range low | CF 14% → 6.6% | Vol.4 p.4 | **J (KEPCO contract price (2026 round-1 cap))** | O · |
| Capacity factor (base) | 16% | Vol.4 p.1 | A | O O |
| KEPCO CfD equity IRR · CAPEX at KEEI top | 8.8% | Vol.4 p.4 | **J (KEPCO contract price (2026 round-1 cap))** | O · |
| Merchant equity IRR · SMP at range top | SMP 148.4 → 8.2% | Vol.4 p.4 | **J (Max gearing · merchant)** | O · |
| Merchant equity IRR · SMP at range low | SMP 90.4 → 0.9% | Vol.4 p.4 | **J (Max gearing · merchant)** | O · |
| SMP base (12-month mainland average) | KRW 113.2 | Vol.4 p.3 | A | O · |
| KTB 3-yr | 3.84% (2026-10-20) | Vol.4 p.A4 | A | O · |
| Merchant equity IRR · 2022 actual unit cost | 7.3% | Vol.4 p.4 | **J (Max gearing · merchant)** | O O |
| KEPCO contract price (2026 round-1 cap) | KRW 147.7 | Vol.4 p.3 | **J (KEPCO contract price (2026 round-1 cap))** | O · |
| Direct PPA price (corridor mid) | KRW 156.5 | Vol.4 p.4 | **J (KEPCO contract price (2026 round-1 cap))** | O O |
| Buyer's room for network charges (per kWh) | KRW 5.6–17.0 | Vol.4 p.5 | **J (Daily output shape exponent)** | O O |
| Seller's required PPA price (per kWh) | KRW 152.8–164.2 | Vol.4 p.i | **J (Daily output shape exponent)** | O O |
| Public example of charges (KEPCO-brokered 3rd-party PPA) | ~KRW 40 | Vol.4 p.5 | B | O O |
| Charge cut (or buyer premium) needed to close | KRW 23 | Vol.4 p.A1 | **J (Daily output shape exponent)** | O · |
| Direct-PPA notice proviso (no grid use) | items 2 and 4 drop out if the T&D grid is not used | Vol.4 p.A4 | A | O · |
| REC article lapses (RE Act addendum art. 2) | end-2029 | Vol.4 p.2 | A | O · |
| Transitional rule for existing REC holders and recognised licensees (addendum art. 3(2)–(3)) | former rules kept | Vol.4 p.2 | A | O · |
| REC revenue for post-2027 entrants | no RECs | Vol.4 p.2 | A | O · |

Source: *1 KSJ Research Vol.1 (Oct 2026) — DART/KRX market-cap split
*2 KSJ Research Vol.2 (Oct 2026) — Korean 40 MW data-centre PF
*3 KSJ Research Vol.3 (Oct 2026) — Target Co. price range
*4 KSJ Research Vol.4 (Oct 2026) — hypothetical 35 MW solar equity

Note: Check: first O = the same value appears on that page of the volume (Korean text); second O = also in the volume's build summary (· = not applicable). Both verified in code at build. The page may show a rounded figure (e.g. 12.25% → 12.2%; exact values in the build summary). Grade = weakest main input (computed values at least D); brackets name that input

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# Glossary: the words on the map

## Prices KRW/kWh · amounts KRW 100mn · ratios % · multiples x

### Glossary

| Term | Definition |
|---|---|
| Invest · wait · avoid | The three verdicts (judgement, J), set by return, closing conditions and flipping assumptions (rule on p2) |
| Trigger | An observable signal that would change a verdict (judgement) |
| Flipping assumption | The input, and the move in it, that changes the conclusion — from each volume's sensitivities |
| Source grade | A official · D computed from official data · B press/external report · C secondary (provisional) · J judgement |
| CfD | Contract for difference with KEPCO; from 2027 the default for new solar |
| PPA | Power purchase agreement — seller and buyer fix price, term and volume directly |
| SMP · REC | System marginal price (wholesale power) · renewable certificate (issuance deadline 31 Dec 2026; existing holders and recognised licensees keep the former rules — RE Act addendum art. 3) |
| DSCR | Debt service coverage = cash flow available ÷ debt service; lenders watch the minimum against a target |
| Condition precedent (CP) | A condition to be met before drawdown or closing |
| Equity IRR · hurdle | Return on equity · minimum return required (judgement in Vol.4) |
| Price range · discount | IPO price range · 1 − range ÷ value per share |
| Double peak · required discount | Risk that earnings and multiples peak together · discount needed to avoid a loss in the 1-year value distribution (P50) |
| Multiple contribution | Share of Δln(cap) due to Δln(P/E) — depends on the earnings basis (NI or OP) |
| Network and ancillary charges | T&D use charges, trading fee, settlements and fund levy a direct-PPA buyer pays |

Note: Shaded = terms new in this volume; the rest match the Vol.1–4 glossaries

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# Afterword

This series began with a single question: "If AI uses this much electricity, where does the money flow, and what does finance have to decide at each turn along the way?"

I studied in the Division of Environmental Science and Ecological Engineering at Korea University, and I am now studying energy and climate at the KAIST Graduate School of Green Growth and Sustainability. The more I studied, the more I found myself curious not about the technology itself but about how it actually gets built. Even with good technology and good policy, nothing gets built unless capital moves. I wanted to understand how projects raise money and how they are structured to earn a return, so I have studied finance on the side.

As an undergraduate I sought out courses in the Department of Food and Resource Economics, and at KAIST I learned accounting principles and climate finance, estimating the cash flows of a biochar plant with financial-engineering methods. In a Hanwha finance competition I priced a product built on a barrier-option structure, and in a project with the ECM division of iM Securities I practised putting what I had learned into the language of practice. This series is the first time I have tried to weave those separate threads of study together on a single industry.

Across the four volumes I tried to sit in a different seat each time. As a sector analyst I took apart a rise in share prices; as a PF lender I set a loan limit; as an ECM banker I drew an IPO price range; and as an equity investor I weighed the price of a power contract. The same industry raised different questions depending on where I sat, and I learned that judgement only becomes solid when those questions are joined together.

I took particular care with the numbers. Every input carries a source and a reliability grade, and I went back to the tariff tables and official notices themselves to check them line by line. After publishing, independent reviews still found assumptions that changed conclusions; I corrected them and kept the corrections on the record rather than hiding them. AI helped me write the code, but deciding what to ask, and checking whether the numbers could be trusted, remained my job to the end.

Even so, this is work I studied and estimated on my own, in the gaps between preparing to graduate and looking for a job. It is an exercise built on public information and assumptions, and to a practitioner's eye it will fall short in many places. I also know that a real deal holds far more: a single clause in a contract, the mood at the negotiating table, risks that numbers do not capture. If you point out where it falls short, I will be grateful to learn from it.

Looking back, I have always tried not to let go of my dreams. While I crossed the boundaries of my major to study finance, there were days when I asked myself whether I had been chasing ideals too much. Honestly, I am not yet sure that I can settle in and grow well as a finance professional. Still, one thing became clear as I finished this series: all that is left now is to move forward.

I want to keep practising the approach of this series. Looking at the same asset, I will think about how someone who manages capital would protect its value, and on what terms someone who entrusts capital would commit it. I will also ask what someone who spots new business opportunities early would look for, and what someone who sets an investment strategy to grow value would change. I believe the starting point for all of this is structuring deals and designing how they are financed. Wherever I sit, my goal is to become a finance professional who judges energy and infrastructure assets through numbers and can explain the basis for that judgement all the way through.

I leave this series not as a finished piece but as the starting point of that path. After my master's, I will hold on to these questions and keep studying.

Thank you to everyone who has read this series.

**Shinjae Kang**

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- This is a personal portfolio for education and research. It is not investment advice or a view on any asset or security.
- Invest / wait / avoid verdicts and triggers are the author's judgement (J). Numbers are carried over from Vol.1–4 with their volume, page and source grade (A2).
- Target Co., the 35 MW solar asset and the data centres are hypothetical. Listed power-equipment names are discussed at sector level, not as a view on any stock.
- Views are the author's own and do not represent any current or future employer.
- Vol.5 as of 2026-10-28 (Vol.1–4 numbers keep their own as-of dates). Methods and code can be reproduced from the repository.
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https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

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